

Ana-Maria Dumitriu

- Romania
- +40773703294
- nmrDumitriu@gmail.com

Software Developer: Full-Stack · AI · Embedded Systems |

Link: <https://ana-dumitriu-portfolio.vercel.app/>

Summary

Engineering student and software developer focused on full-stack and AI-powered applications. Hands-on experience building and deploying end-to-end products using TypeScript, React/Next.js, Python/FastAPI, REST APIs, PostgreSQL and Git. Built systems involving authentication, session management, relational databases, responsive interfaces and AI integration. Strong problem-solving background across software, embedded systems and technical product development.

Skills

- **Languages:** Python · TypeScript · JavaScript · Java · C/C++
- **Frontend:** React · Next.js · React Native · HTML · CSS · Responsive Design
- **Backend & APIs:** FastAPI · REST APIs · Authentication · Session Management
- **Databases:** PostgreSQL · SQLite · SQL · Alembic
- **AI:** LLM Integration · Prompt Engineering · AI-assisted Development
- **Tools & Deployment:** Git · GitHub · Vercel · Render
- **Embedded:** ESP32 · Arduino · BLE · Sensors · Motor Control
- **3D/CAD:** Fusion 360 · Blender · AutoCAD · FDM Printing

Projects

AutoDiagnose AI — Full-Stack AI Automotive Diagnostic Platform

TypeScript · Next.js · React · Python · FastAPI · PostgreSQL · REST APIs · AI

- Built and deployed an end-to-end full-stack application with responsive desktop/mobile interfaces, reusable React components and REST API integration.
- Developed the FastAPI backend, multi-user authentication, session management, relational data model and persistent user-specific diagnostic cases.
- Designed a diagnostic engine combining vehicle data, symptoms and OBD-II/DTC codes with confidence-based reasoning, with architecture prepared for LLM/NLP integration.
- Debugged production issues across frontend, API, authentication, database and deployment layers.

GitHub: github.com/DumitriuAna23/autodiagnose-ai

ELIO — AI Educational Robot & Connected Software System

C/C++ · ESP32 · React Native · TypeScript · BLE · AI · Fusion 360

- Designed and developed an intelligent educational robot integrating embedded C/C++ firmware, ESP32 hardware, sensors, motor control, BLE communication and real-time telemetry.
- Developed a companion mobile application for robot control, AI-powered interaction and educational activities, including visual block-based programming inspired by Scratch.
- Integrated software and hardware components and troubleshoot communication, control and system-level issues throughout development.

GitHub: github.com/DumitriuAna23/elio-ai-robot

Experience

Assistant Manager & Technical Trainer

Tales & Tech HUB | Suceava | November 2025 - Current

- Contribute to technical software and embedded projects involving interactive applications, ESP32 systems, sensors and custom hardware.
 - Troubleshoot software, code, electronics and hardware issues during technical project development.
 - Deliver hands-on workshops in programming, C/C++, Arduino, robotics and emerging technologies.
 - Support technical documentation, planning, communication, reporting and project coordination.
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Education

B.Sc. in Electrical Engineering & Computer Science: Automotive Command and Control Systems

Ștefan cel Mare University of Suceava | October 2024 - Current

- Third-year student in a four-year Bachelor's program.
 - Relevant coursework: Computer Programming, Artificial Intelligence, Microprocessors & Microcontrollers, Applied Electronics, Automotive Control Systems.
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Languages

Romanian — Native

English — Upper-Intermediate (B2)